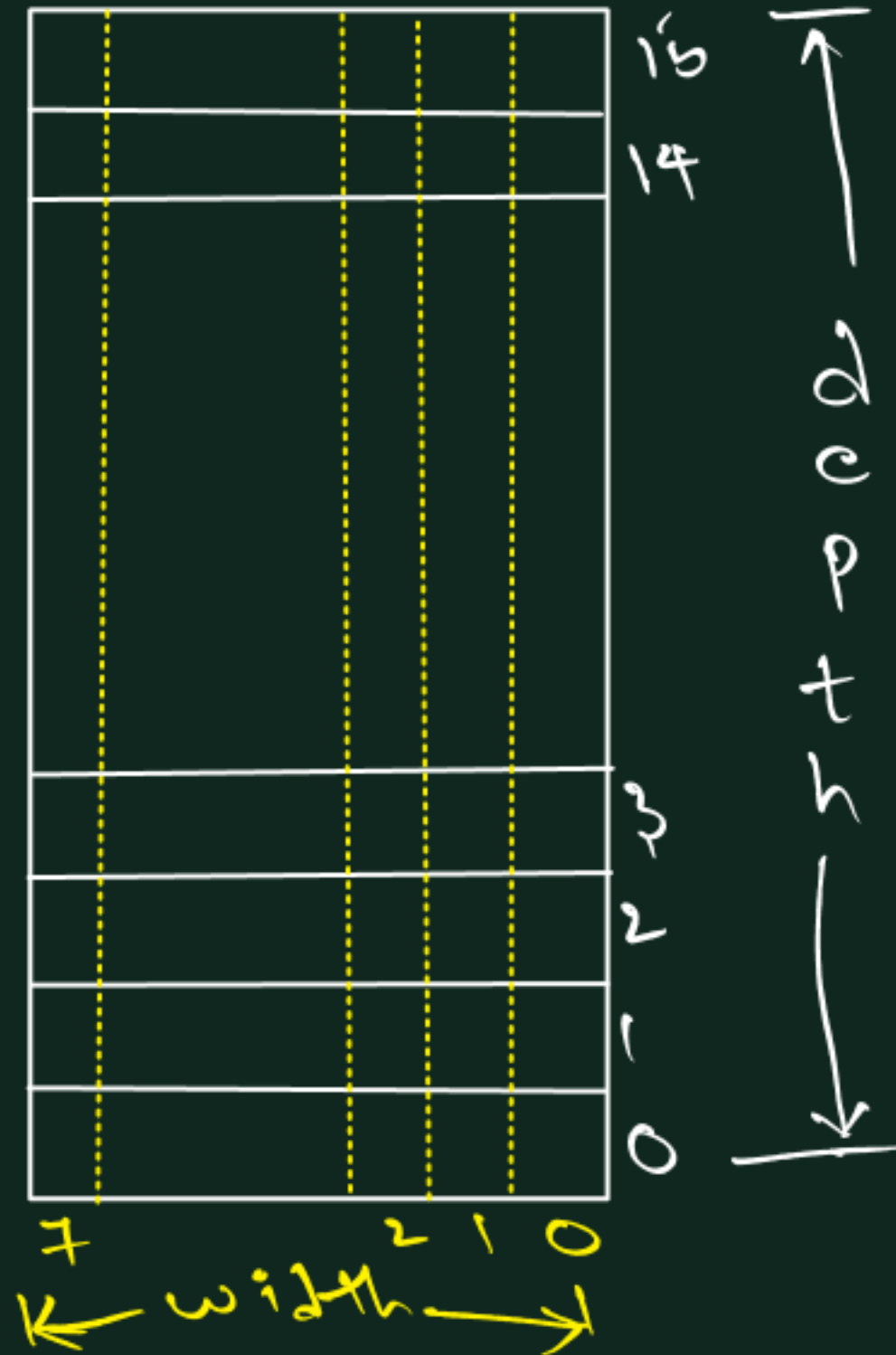


→ reg[7:0] mem[15:0]



reg[width-1:0] mem[depth-1:0];

Size = width * depth
Size = 8 * 16
⇒ 128 bits of data

$$\text{Size} = 250 \text{ mb} \mid \text{width} = 512 \mid \text{depth} = ?$$

$$\text{depth} = \frac{\text{Size}}{\text{width}} \Rightarrow \frac{250 \times 2^{26}}{2^9} \Rightarrow 250 \times 2^{17}$$

$$\Rightarrow 512000$$

$$\underline{256 \times 1024 \times 1024}$$

$$\Rightarrow \frac{2^8 \times 2^{20}}{2^9} \Rightarrow \frac{2^{28}}{2^9} \Rightarrow \frac{2^{19}}{2^9} \Rightarrow \underline{524288}$$

$$\text{reg}[511:0] \text{ mem}[52487:0]; \rightarrow \textcircled{1}$$

$$\text{reg}[2^{**9}-1:0] \text{ mem}[2^{**19}-1:0]; \rightarrow \textcircled{2}$$

$$\text{Size} = 16\text{B} \mid \text{Depth} = 512 \mid \text{width} = ? \quad (2097151)X$$

$$\Rightarrow \frac{2^{30} \times 2^3}{2^9} \Rightarrow \frac{2^{33}}{2^9} \Rightarrow \underline{\binom{2^4}{2}} (16777216)$$

$$\checkmark \text{reg}[2^{**24}-1:0] \text{ mem}[2^{**9}-1:0]$$

Size = 16b | width = 32 bits | depth = ? (33554432)

$$\Rightarrow \frac{2^{\cancel{30}}}{2^{\cancel{5}}} \Rightarrow 2^{25} \Rightarrow \text{reg}[2^{**}5-1:0] \text{ mem}[2^{**}25-1:0]$$

$$\frac{1024 * 1024 * 1024}{32} \Rightarrow$$

Size = 1 Kb depth = 32 loc

$$\text{width} = ? \quad \frac{2^{\cancel{10}}}{2^{\cancel{5}}} \Rightarrow 2^5$$

$$\Rightarrow 32 \quad \text{reg}[31:0] \text{ mem}[31:0];$$